

Vittorio Rossi

AI Engineer & Researcher

rossi.vittori@gmail.com | +39 3458367454 | vittoriorossi.com | linkedin.com/in/vitrossi | github.com/VittorioRossi | Google Scholar

RESEARCH & PUBLICATIONS

Diffusion Language Models Are Natively Length-Aware Accepted, NeurIPS 2026 Workshop on Diffusion Language Models

Vittorio Rossi, Giacomo Cirò, Davide Beltrame, Luca Gandolfi, Paul Röttger, Dirk Hovy

- **First author.** Showed that masked diffusion LLMs encode the required output length before decoding, and turned that into a **zero-shot, training-free method** shortening the generation canvas from a single forward pass
- **>30% FLOPs reduction on every benchmark, up to 90% on some**, with no statistically significant performance loss, and **improved performance on 2 of 4** (GSM8K, HumanEval, IfEval, LongFormQA on LLaDA-8B)
- Ran the experimental program on **H100 GPUs via SLURM**; owned hypothesis through manuscript

EDUCATION

Bocconi University

September 2024 – December 2026

MSc in Artificial Intelligence, exam average 28/30

Milan, Italy

Georgia Institute of Technology

January 2024 – May 2024

Exchange Programme

Atlanta, USA

Bocconi University

September 2021 – July 2024

BSc in Economics, Management and Computer Science, 110/110

Milan, Italy

PROFESSIONAL EXPERIENCE

Commerce Clarity

June 2024 – Present

Founding AI Engineer

Milan, Italy

- Joined pre-team as the **first AI hire**; built and own the **workflow execution engine** running the company's product-enrichment, translation and data-sourcing agent pipelines
- Built the **LLM infrastructure layer** for prompt versioning, multi-provider clients and execution analytics, tracking **10M+ production calls**
- Fine-tuned a **Gemma 3** model with **LoRA** on a MacBook to adapt tone of voice for a client
- Built the pipeline generating a **1M-product catalog**, plus a vision pipeline matching 10K products against 5M competitor listings
- Defined the company's **evaluation standard**: a standardized eval framework setting out what evaluation means in this context, with rubrics and a fixed spec format reused across customer workflows
- Built the **evaluation harnesses** behind it: an LLM-as-judge scorer and a general harness running the framework across customer workflows

TECHNICAL PROJECTS

IRE (Integrated Research Environment)

May 2026 – Present

Co-founder

- **Co-founded and released** an open-source, local-first research environment for ML researchers, built in **Rust** and **TypeScript**, with macOS and Linux releases under the MIT license
- Designed a pluggable agent-provider interface unifying Claude Code, Codex and OpenCode, and a native MCP bridge giving agents a Git-tracked wiki, structured memory and non-blocking experiment execution

GraphCode-Bench

March 2026

LLM Evaluation Benchmark

- Built a benchmark for agentic code understanding over graph-structured repository representations, comparing **five open-weight models**: Qwen3 14B, Qwen3-Coder 30B, GPT-OSS 20B, Mistral-Small 24B, and Llama-3.1 8B

GraphyDocs

January 2025 – March 2026

Codebase Knowledge Graphs for LLM Retrieval

- Built a full-stack system converting codebases into **structured knowledge graphs** for LLM retrieval, addressing context limits on large repositories, with a Neo4j backend, FastAPI service and React front end

TECHNICAL SKILLS

Languages

Python (expert, Pandas, NumPy), TypeScript, Rust, SQL, PHP, C/C++

ML and LLM Systems

PyTorch, Transformers, LoRA, diffusion LMs, inference optimisation, evaluation and benchmarking

Infrastructure and Data

SLURM and HPC (H100), Kubernetes, Docker, Google Cloud Platform, CI/CD, Neo4j, MongoDB

Spoken

Italian (native), English (fluent, IELTS 7.5), Spanish (basic)